

Final Report for Neo Coffee shop dataset



Data Preparation

The preparation phase converted 1,000 raw transaction records into a stable format for analysis. The process focused on two critical areas: handling missing contact information and neutralizing extreme price fluctuations to ensure statistical reliability.

1. Handling Missing Data (Categorical)

Audit results showed gaps in the Customer Email and Phone Number fields.

- **Action:** All null entries were filled with the constant string "Unknown".
- **Reasoning:** Since personal identifiers cannot be calculated using averages or medians, a placeholder was necessary to keep the dataset size intact (\$N=1000\$).
- **Outcome:** This prevents the loss of sales data associated with those rows and allows for future comparisons between "Contactable" vs. "Anonymous" customer spending habits without breaking data type schemas.

2. Outlier Management (Numerical)

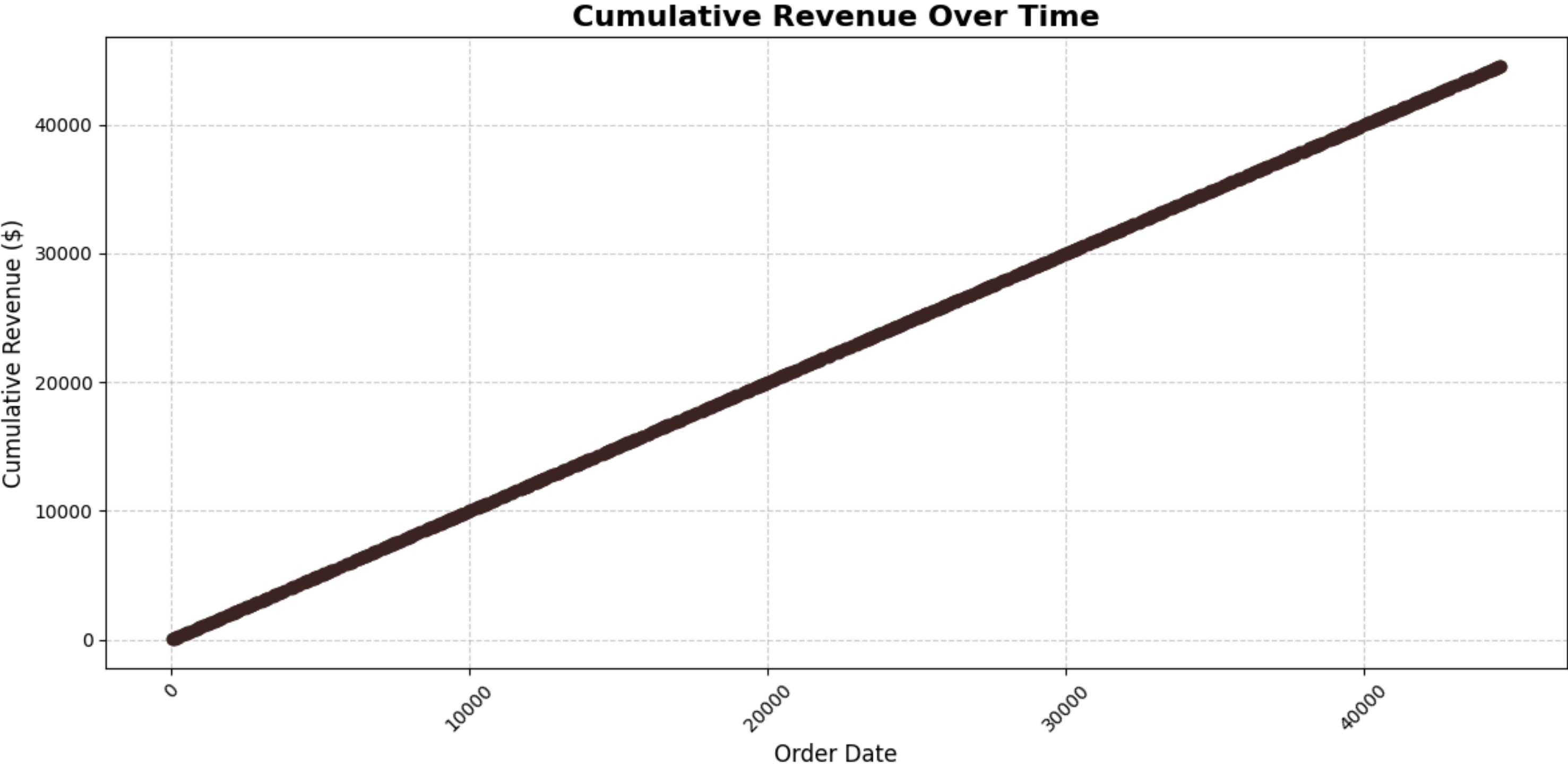
To stabilize the mean and variance, outliers in Unit Price, Price per 100g, and Profit were addressed using the Interquartile Range (IQR) method.

- **Action:** Applied Winsorization (Capping). Values exceeding the statistical boundaries ($Q1 - 1.5 \times IQR$ or $Q3 + 1.5 \times IQR$) were reset to those limit values using the NumPy clip function.
- **Reasoning:** Extreme values—likely representing bulk orders or rare specialty roasts—are valid revenue but can distort averages.
- **Outcome:** Capping retains the revenue within the dataset but prevents a few large sales from skewing the overall performance metrics.

3. Post-Cleaning Status

- **Missing Values:** 0 remaining.
- **Data Shape:** The distribution is now compact and ready for inferential testing.
- **Verification:** Boxplot analysis confirms that tails have been pulled in to the maximum "whisker" limits, stabilizing the standard deviation for more accurate forecasting.

Cumulative Revenue Trend



Cumulative Revenue Growth Trend

1. Revenue Trajectory Performance

The cumulative revenue analysis (2021–2024) confirms a linear and stable growth pattern. The absence of volatility, plateaus, or sharp spikes indicates that Neo Coffee Shop operates with high financial predictability. This "smooth" progression signifies a business that has moved beyond the erratic startup phase into a matured, reliable accumulation cycle.

2. Core Drivers of Stability

The observed trend is likely maintained by a combination of internal and external factors:

- **Customer Retention:** A stable baseline of repeat business provides a "floor" for daily revenue, preventing the sharp dips typically seen in seasonal or trend-dependent businesses.
- **Market Alignment:** The product mix and pricing strategy are well-calibrated to local demand, ensuring consistent market acceptance over the three-year period.
- **Operational Resilience:** The lack of stagnation suggests the business has successfully navigated external headwinds—such as supply chain shifts or economic fluctuations—without compromising sales volume.

3. Business Health Assessment

Steady linear growth is a primary indicator of a sustainable business model.

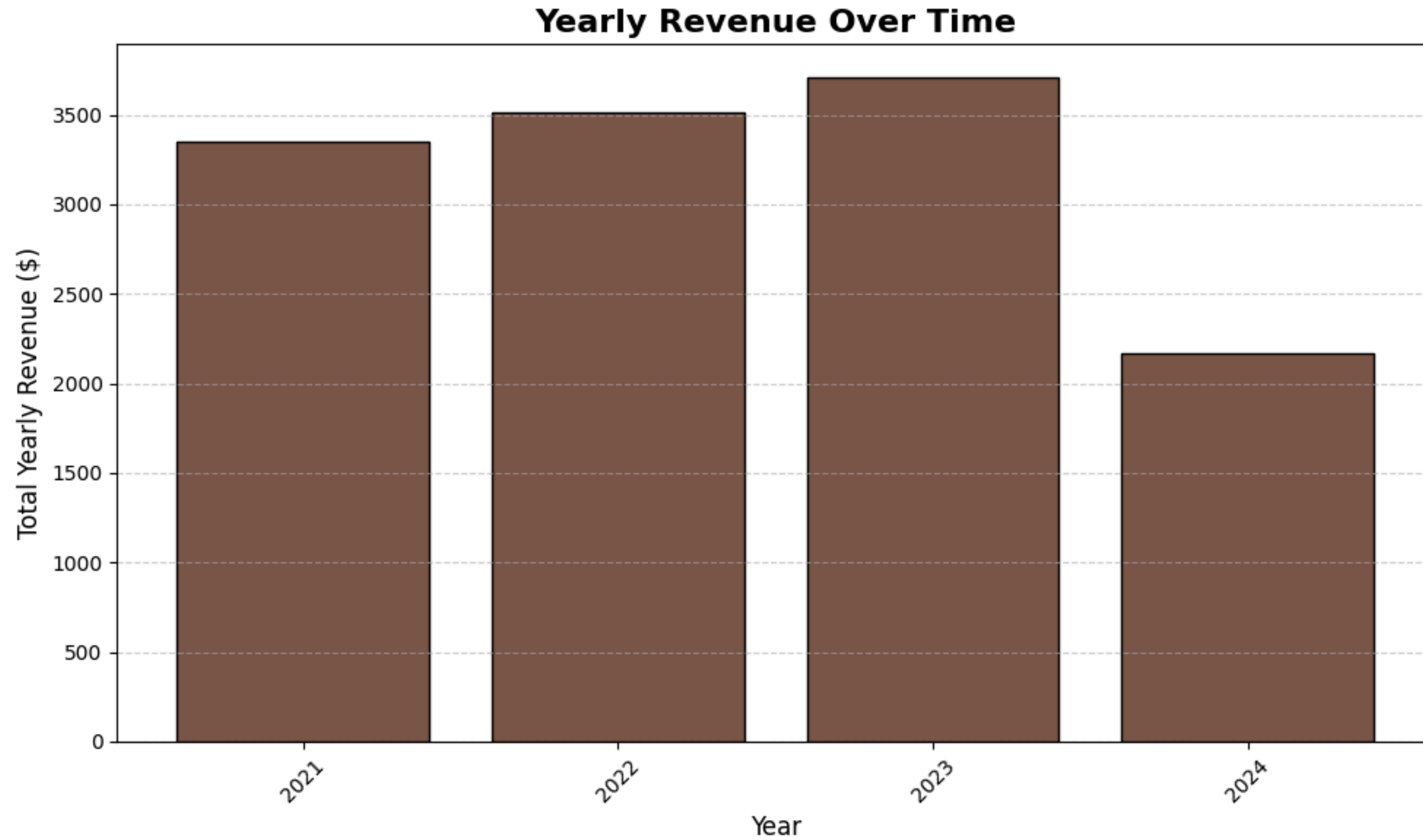
- **Financial Planning:** Predictable cash flow facilitates lower-risk reinvestment and long-term capital planning.
- **Operational Maturity:** The steady climb reflects an efficient supply chain and staffing model capable of supporting consistent throughput.
- **Stakeholder Confidence:** From an analytical maturity standpoint, this trend represents a low-risk profile, making the business an attractive candidate for expansion or external investment.

4. Strategic Growth Accelerants

To evolve from steady growth to accelerated growth, the following data-driven strategies are recommended:

- **Refinement of Loyalty Framework:** Move beyond basic purchase tracking to Personalized Incentives. Use transaction history to offer rewards tailored to individual taste preferences (e.g., targeting Arabica drinkers with exclusive early access to new specialty roasts).
- **Optimization of Average Order Value (AOV):** Implement "Smart Bundling" and seasonal innovation. Introducing high-margin, limited-time food or merchandise items can increase the revenue per transaction without requiring new customer acquisition.
- **Data-Driven Expansion:** Analyze geographic sales data to identify clusters of high-value customers. This insight should dictate the location of new physical branches or targeted digital marketing zones.
- **Dynamic Pricing Models:** Evaluate price elasticity by variety. Given the stability of the current model, the shop may have the "pricing power" to marginally increase prices on premium varieties (like Liberica) to accelerate the slope of the revenue line.

Yearly Revenue Trend



Multi-Year Revenue Performance

1. Yearly Revenue Trajectory (2021–2024)

The financial data highlights a period of sustained expansion followed by a partial-year reporting phase.

- **2021:** Base year established with a revenue of \$3,351.35.
- **2022:** Grew to \$3,510.06, confirming successful early-stage market penetration.
- **2023:** Continued momentum, reaching \$3,707.25.
- **2024** (Incomplete): Current year-to-date stands at \$2,165.51.

2. Analytical Context of the 2024 Figures

The apparent decline in 2024 is not an indicator of poor performance, but a result of incomplete data reporting. As the dataset likely concludes mid-year, the \$2,165.51 figure represents only a portion of the fiscal cycle. When normalized against the same periods in previous years, 2024 is likely on track to meet or exceed 2023 levels.

3. Strategic Significance of Historical Growth

The 2021–2023 trend provides high-level proof of a scalable business model:

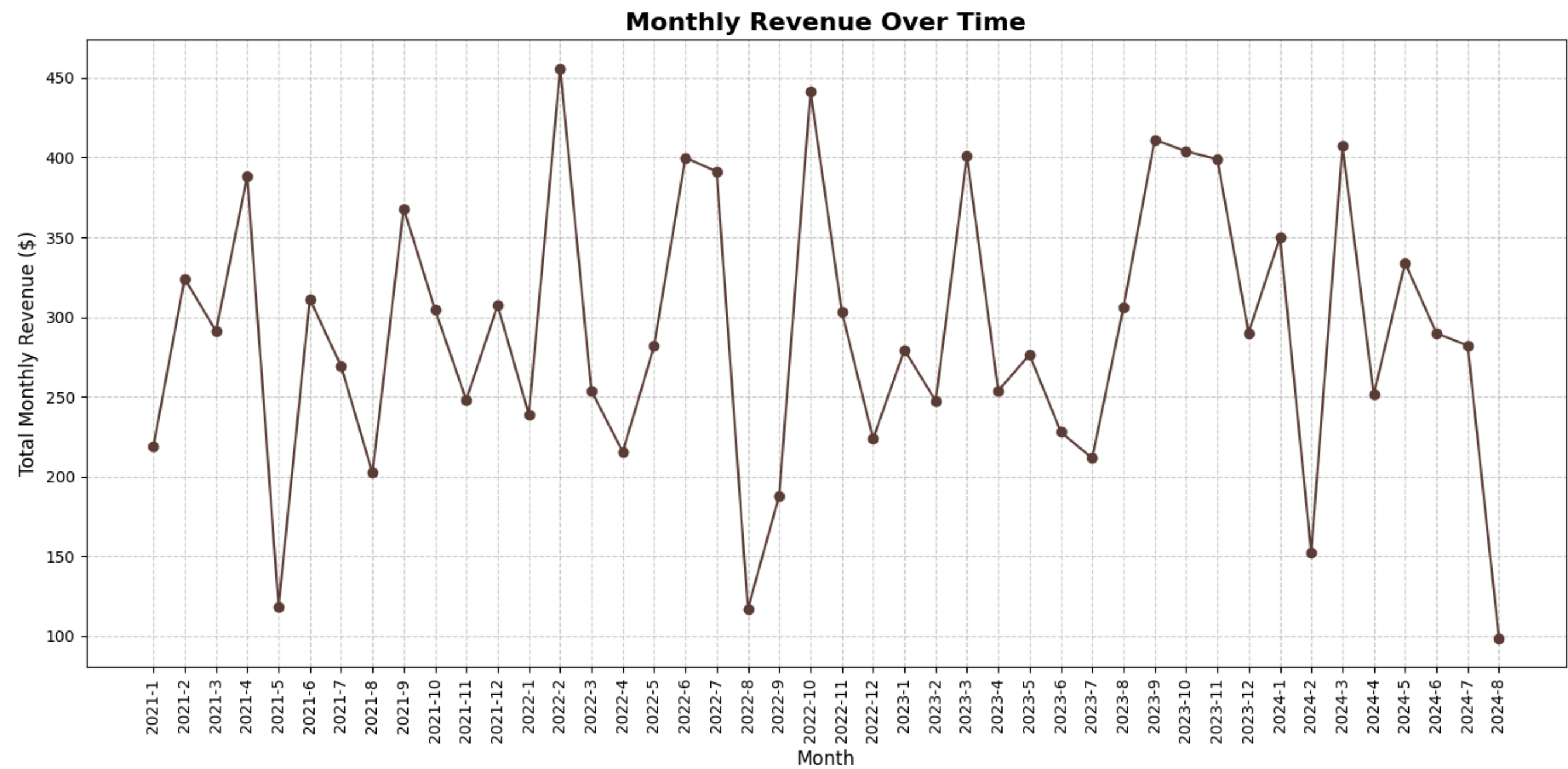
- **Proof of Concept:** Consistent growth over three years indicates that the shop has moved past the "novelty" phase and has established a loyal customer base.
- **Investment Readiness:** This trajectory is a primary metric for stakeholders, demonstrating that the business can effectively turn operational capital into increasing returns.
- **Predictive Baseline:** The 10.6% total growth between 2021 and 2023 serves as a reliable benchmark for setting 2025 targets and determining inventory procurement budgets.

4. Future Planning and Monitoring Objectives

To maximize the momentum identified in the 2021–2023 period, the following actions are prioritized:

- **Year-End Projections:** Use the monthly average from the first half of 2024 to project the final year-end total. This allows for proactive budgeting for the Q4 holiday peak.
- **Growth Factor Identification:** Isolate whether the \$350+ increase between 2021 and 2023 came from price increases, new customer acquisition, or higher purchase frequency from existing loyalty members.
- **Seasonality Calibration:** Cross-reference the yearly growth with monthly fluctuations to determine if certain months are growing faster than others. For example, if "Mid-Year Dips" are becoming less severe each year, it indicates successful contracyclical marketing.
- **Targeted Scaling:** With three years of verified growth, the business is now in a position to evaluate scaling opportunities, such as increasing the production of high-margin varieties (Liberica/Excelsa) to further boost the 2025 revenue ceiling.

Monthly Revenue Trend



Monthly Revenue and Seasonal Dynamics

1. Trend and Volatility Overview

The monthly revenue data from January 2021 to August 2024 reveals a business characterized by high intra-year volatility despite a positive long-term trajectory. While the cumulative view shows stability, the monthly view highlights a non-linear path where revenue is subject to significant short-term shifts.

2. Seasonality and Cyclical Patterns

The analysis identifies recurring "Revenue Valleys" and "Growth Peaks" that suggest a seasonal heartbeat influenced by external factors:

- **Mid-Year Resistance:** Recurring dips in May and August indicate a cooling period. This could be attributed to warmer weather reducing the demand for hot beverages or shifts in local foot traffic during summer vacation periods.
- **Late/Early-Year Momentum:** Peaks in November, February, and March align with colder weather and major holiday seasons, which typically drive higher consumption of premium roasts and gift-based purchases.
- **Inconsistent Spikes:** The lack of perfectly identical year-over-year peaks suggests that while seasonality exists, revenue is also heavily driven by event-based variables such as specific marketing campaigns or local economic shifts.

3. Operational and Financial Implications

The presence of these fluctuations requires a move away from "flat" operational planning toward Dynamic Resource Allocation:

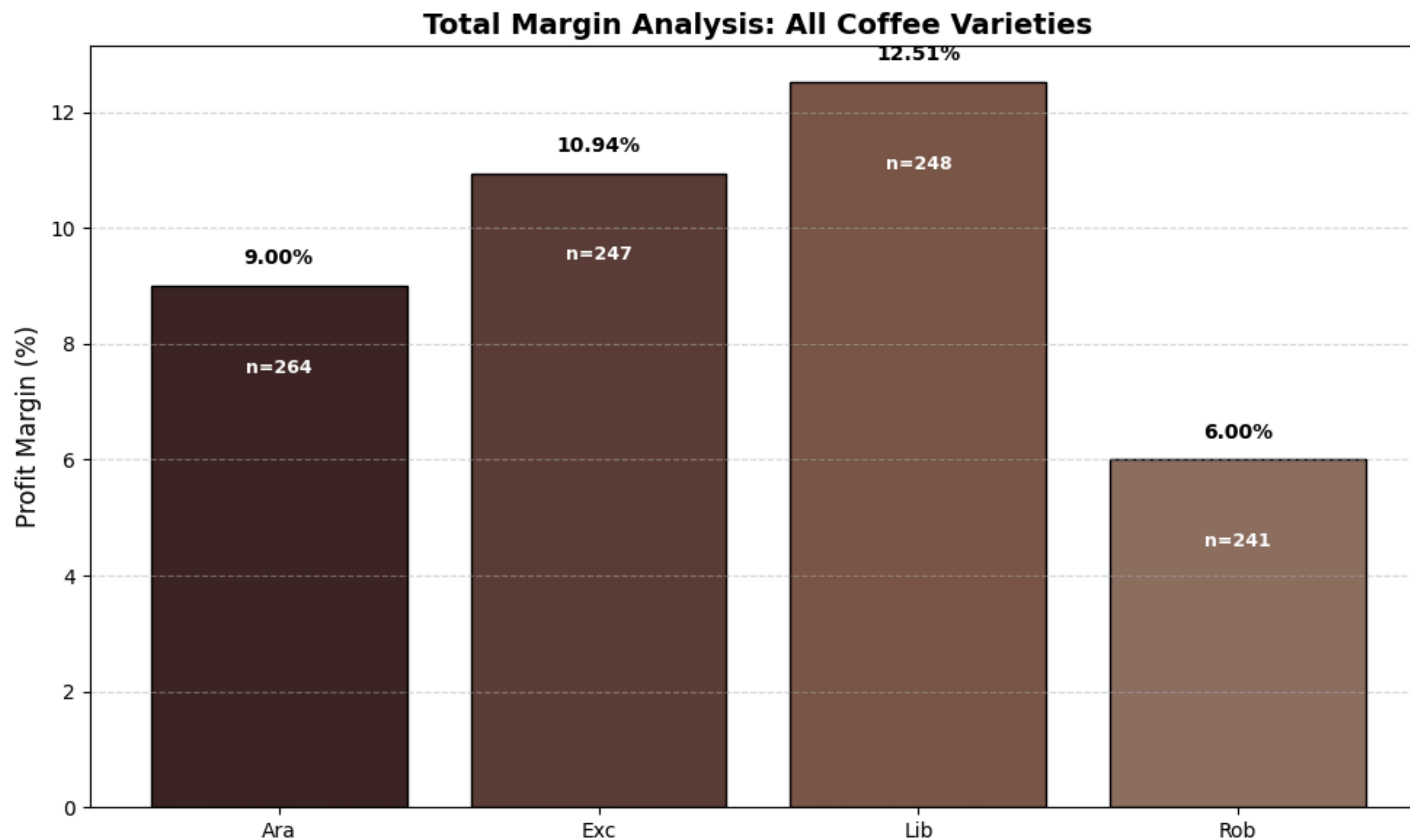
- **Inventory Tuning:** To prevent capital being tied up in slow-moving stock, procurement must be tightened in April and July to prepare for the May/August dips.
- **Labor Efficiency:** A flexible staffing model is essential. Overstaffing during identified "trough" months directly erodes the net margin, while understaffing during peaks leads to lost sales through long wait times.
- **Cash Flow Buffering:** Financial forecasting must account for these cycles to ensure that the "Peak" months provide enough of a cushion to sustain operations during the "Trough" periods.

4. Targeted Strategic Actions

To flatten the valleys and maximize the peaks, the following tactical shifts are recommended:

- **Contracyclical Promotions:** Launch high-impact campaigns specifically in May and August. This could include "Summer Iced Series" launches or "Mid-Year Loyalty Multipliers" to incentivize visits during traditionally slow periods.
- **Weather-Adaptive Product Mix:** Diversify the menu to protect against temperature-related dips. Increasing the visibility of cold brews, nitro coffees, and chilled specialty teas can sustain revenue when hot coffee demand wanes.
- **Event-Synchronized Marketing:** Align promotions with local university calendars or community events. If a "Peak" in March is tied to a local festival, the shop should maximize its presence through pop-ups or event-specific bundles.
- **Predictive Stocking:** Use historical monthly data to automate inventory triggers. This ensures that high-margin beans (like Liberica) are always in stock during the February/March surge without over-ordering in the preceding months.

Coffee Type Margin Analysis



Coffee Type Margin Analysis

All Coffee Varieties" provides a clear overview of the profit margins for different coffee types offered by Neo Coffee Shop: Arabica (Ara), Excelsa (Exc), Liberica (Lib), and Robusta (Rob).

Analysis Summary: Profitability by Variety

The analysis of the four coffee varieties reveals a clear hierarchy in margin efficiency:

- **Top Performers:** Liberica (12.51%) and Excelsa (10.94%) are the most profitable per unit. These figures suggest either a lower cost of goods sold (COGS) or a successful premium pricing strategy based on rarity and flavor profile.
- **Mid-Range:** Arabica (9.00%) maintains a stable margin, likely due to its role as a high-volume, competitive market staple.
- **Underperformer:** Robusta (6.00%) provides the lowest return. This suggests its current role is likely as a price-leader or a blend component rather than a primary profit driver.

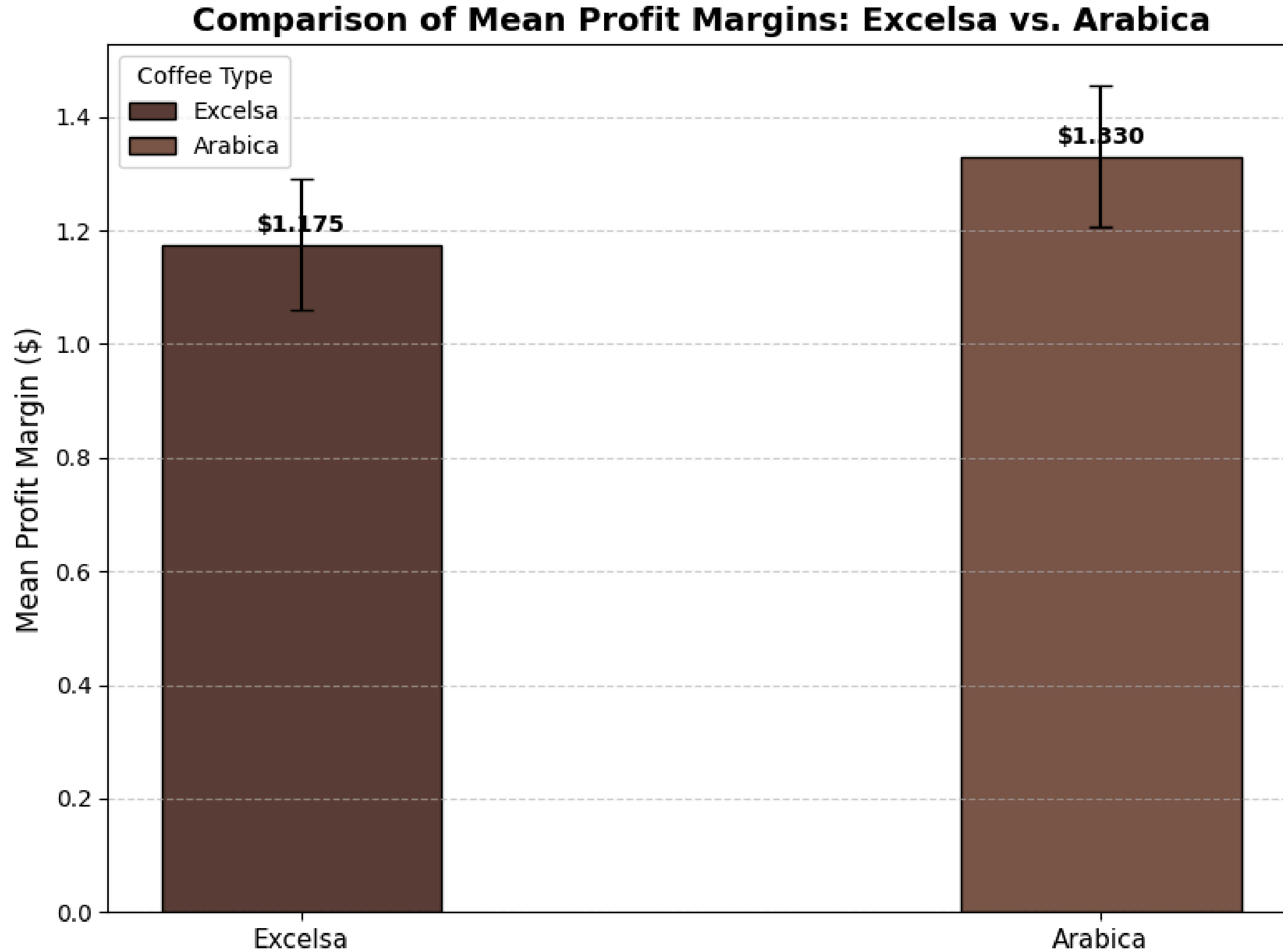
Strategic Implications

1. **Product Mix Shift:** There is a clear financial incentive to move customers "up the value chain" toward Liberica and Excelsa through targeted marketing and increased visibility.
2. **Margin Optimization:** For Arabica, small gains in sourcing efficiency or minor price adjustments could yield significant total profit given its likely high sales volume.
3. **Strategic Review of Robusta:** If Robusta is not essential for customer acquisition or specific blends, its position in the catalog should be re-evaluated to prevent it from diluting the overall shop margin.

Action Plan & Next Steps

- **Cost Audit:** Perform a deep-dive into the procurement, roasting, and packaging costs for each variety to find specific savings, particularly for low-margin items.
- **Price Elasticity Study:** Conduct market research to determine if Liberica and Excelsa can support further price increases and if Robusta can be priced higher without losing volume.
- **Marketing Realignment:** Launch campaigns that emphasize the unique specialty characteristics of the high-margin varieties.
- **Inventory Rebalancing:** Adjust stock levels to prioritize the availability of more profitable beans while minimizing waste in lower-performing categories.

Excelsa vs. Arabica Profit Margins



Comparative Profitability of Excelsa vs. Arabica

1. Comparative Performance Metrics

The statistical analysis evaluated the profit margins of Excelsa and Arabica varieties to determine if observed differences represent a real business trend or mere sampling noise.

- **Excelsa:** Mean profit margin of 1.175 (95% CI: 1.059 to 1.292).
- **Arabica:** Mean profit margin of 1.330 (95% CI: 1.206 to 1.454).

2. Inferential Testing & Significance

While Arabica displays a numerically higher average, the Welch’s t-test indicates that this gap is not definitive.

- **P-value:** 0.074
- **Threshold (α):** 0.05

Since the p-value is greater than the 0.05 significance level, we fail to reject the null hypothesis. The significant overlap in the 95% confidence intervals reinforces this conclusion: the data does not provide enough evidence to claim that one variety is inherently more profitable than the other.

3. Business Implications: Profit Neutrality

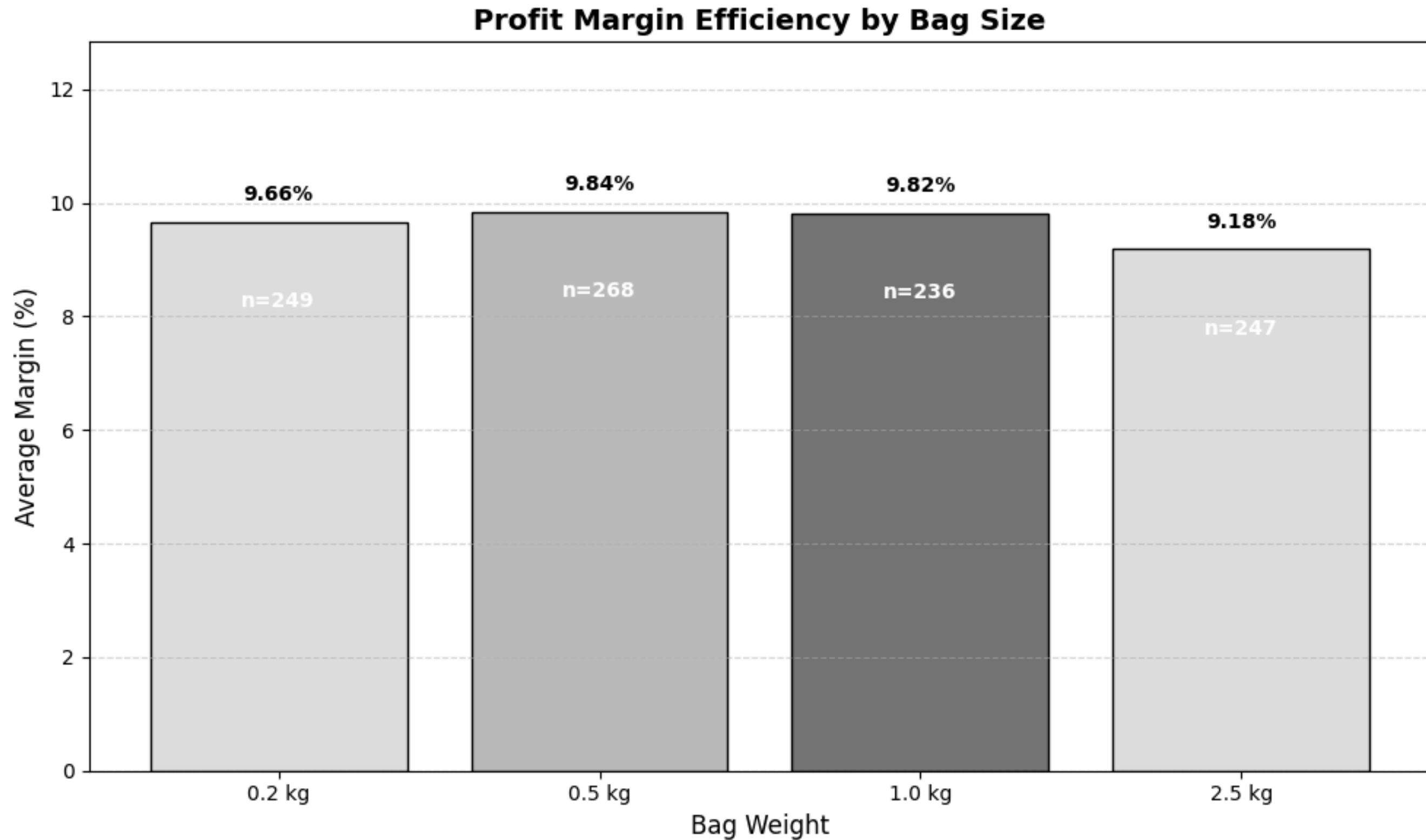
From an analytical perspective, these results suggest Profit Neutrality. Prioritizing inventory or marketing spend solely based on the margin difference between these two types would be statistically premature. Both varieties contribute similarly to the bottom line, meaning the "numerical lead" held by Arabica could fluctuate or disappear with a larger sample size.

4. Strategic Recommendations for Product Differentiation

Because profit margins are not a differentiator, the strategy must pivot to secondary metrics and qualitative value:

- **Volume-Based Prioritization:** Analyze total sales volume (quantity). A variety with a similar margin but double the turnover rate generates significantly more total cash flow.
- **Customer Sentiment & Demand:** Use market research to identify if one variety acts as a "gateway product" that attracts new customers, even if its per-unit profit is identical to others.
- **Cost Structure Audit:** Investigate the supply chain for both types. Since the margins are close, even a small reduction in sourcing or roasting costs for one could push it into a state of statistical superiority.
- **Strategic Positioning:** Treat both as core staples. Focus branding on flavor profiles—positioning one as a "premium specialty" and the other as a "daily reliable"—to capture different market segments without needing to favor one's profitability over the other.

Coffee Bag Size Margin Analysis



Analysis Report: Profit Margin Efficiency by Bag Size

1. Performance Overview

Data from Neo Coffee Shop reveals consistent profit margins across all packaging tiers, with a narrow variance of 0.66 percentage points between the highest and lowest performers.

0.5 kg bag: Highest average margin at 9.84% (\$n=268\$ orders).

1.0 kg bag: Second highest margin at 9.82% (\$n=236\$ orders).

0.2 kg bag: Strong margin at 9.66% (\$n=249\$ orders).

2.5 kg bag: Lowest average margin at 9.18% (\$n=247\$ orders).

2. Analytical Findings

Margin Leaders: The 0.5 kg and 1.0 kg units demonstrate the highest efficiency. These sizes likely balance optimal price points with manageable packaging and storage costs.

Bulk Compression: The 2.5 kg unit exhibits the lowest margin. This suggests that while bulk purchasing creates volume, the current pricing structure likely incorporates discounts to incentivize large-scale buyers, which slightly dilutes the per-unit profitability.

Consumer Behavior: Order volumes are evenly distributed across all sizes (\$n \approx 236-268\$), indicating that customer demand is not dictated by price-per-gram alone, but rather by specific needs such as freshness (smaller bags) or convenience (larger bags).

3. Operational and Strategic Drivers

Pricing Architecture: The data suggests a tiered pricing strategy where per-unit prices decrease as volume increases. This successfully attracts bulk buyers but requires careful monitoring to ensure bulk discounts do not excessively erode margins.

Cost Structures: While packaging larger quantities is often more efficient per gram, the absolute cost of heavier-duty materials and specialized handling for 2.5 kg units may offset some of the anticipated economies of scale.

Value Perception: Smaller bags (0.2 kg and 0.5 kg) likely command a "premium for freshness," allowing the shop to capture higher margins from customers testing new roasts or prioritizing storage space.

4. Strategic Recommendations

Inventory Prioritization: Maintain high stock levels for 0.5 kg and 1.0 kg units to capitalize on their superior margin efficiency and high order frequency.

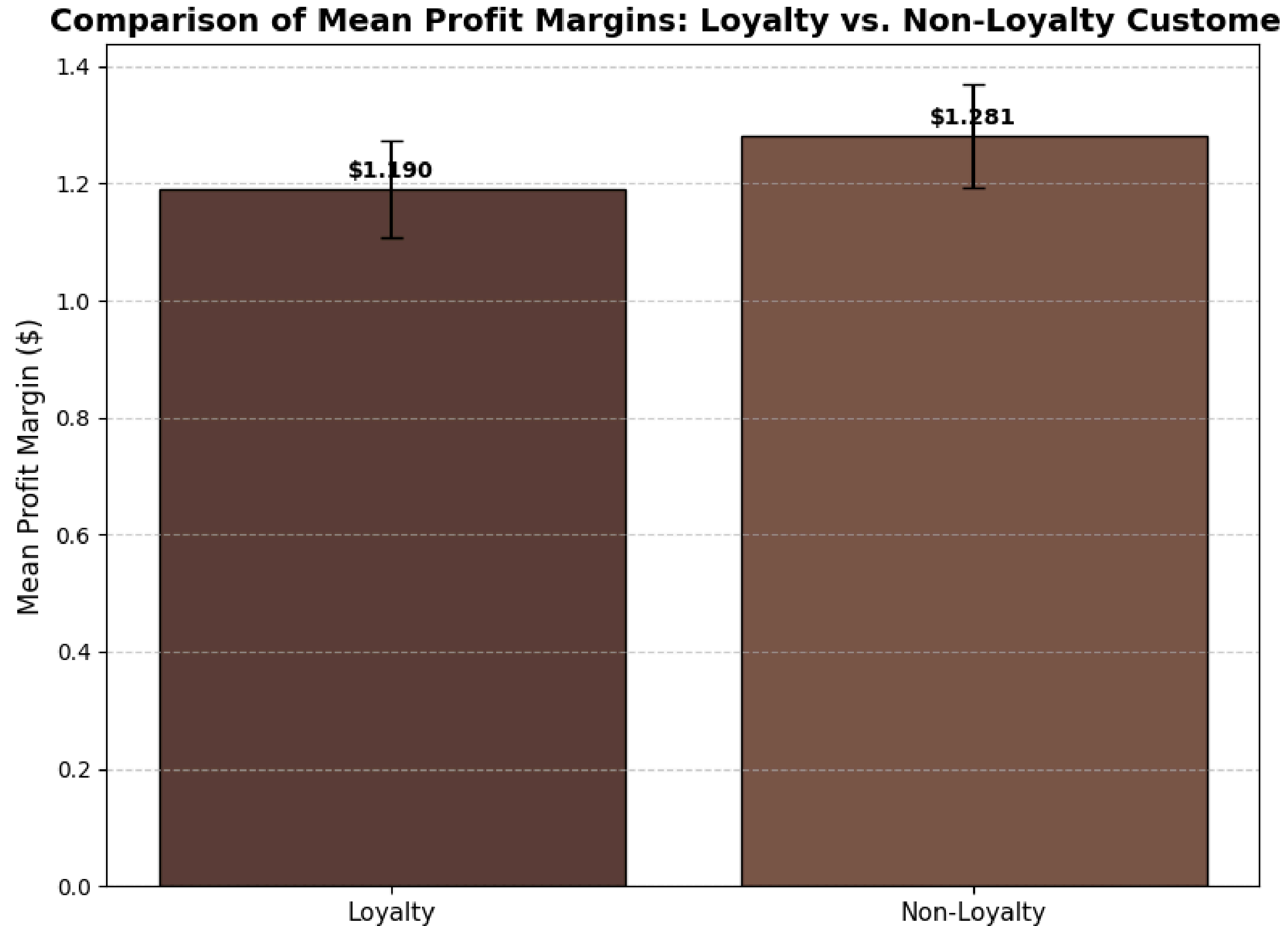
Bulk Margin Optimization: Conduct a price sensitivity analysis on the 2.5 kg unit. A marginal price adjustment could improve profitability without deterring bulk customers. Alternatively, replace direct discounts with value-adds (e.g., loyalty multipliers) to preserve the unit price.

Promotion of Mid-Tier Sizes: Focus marketing efforts on 0.5 kg and 1.0 kg offerings. These sizes represent the "sweet spot" of the product lineup, providing the best return on investment.

Packaging Audit: Review supplier agreements and material costs for the 2.5 kg bags to identify opportunities for reducing the absolute cost of bulk packaging.

Market Expansion: Explore seasonal or limited-run roasts exclusively in high-margin 0.2 kg and 0.5 kg formats to leverage impulse buying and perceived scarcity.

Loyalty vs. Non-Loyalty Profit Margins



Impact of Loyalty Program on Profit Margins

1. Comparative Performance Metrics

The analysis compared the transactional profit margins of customers enrolled in the loyalty program against those who are not.

- **Loyalty Customers:** Mean profit margin of \$1.190 (95% CI: \$1.107 to \$1.273).
- **Non-Loyalty Customers:** Mean profit margin of \$1.281 (95% CI: \$1.194 to \$1.369).

2. Inferential Testing Results

An independent samples t-test was conducted to determine if the \$0.091 numerical difference in favor of non-loyalty customers was statistically significant.

- **T-statistic:** -1.487
- **P-value:** 0.137
- **Significance Level (\$\alpha\$):** 0.05

Since the p-value (0.137) exceeds the 0.05 threshold, the results fail to reject the null hypothesis. The overlapping confidence intervals and the p-value indicate that the observed difference is likely due to random sampling variation rather than a fundamental difference in spending behavior between the two groups.

3. Strategic Implications for the Loyalty Program

The findings reveal that the loyalty program, as currently structured, does not provide a statistically significant "lift" to the profit margin per transaction.

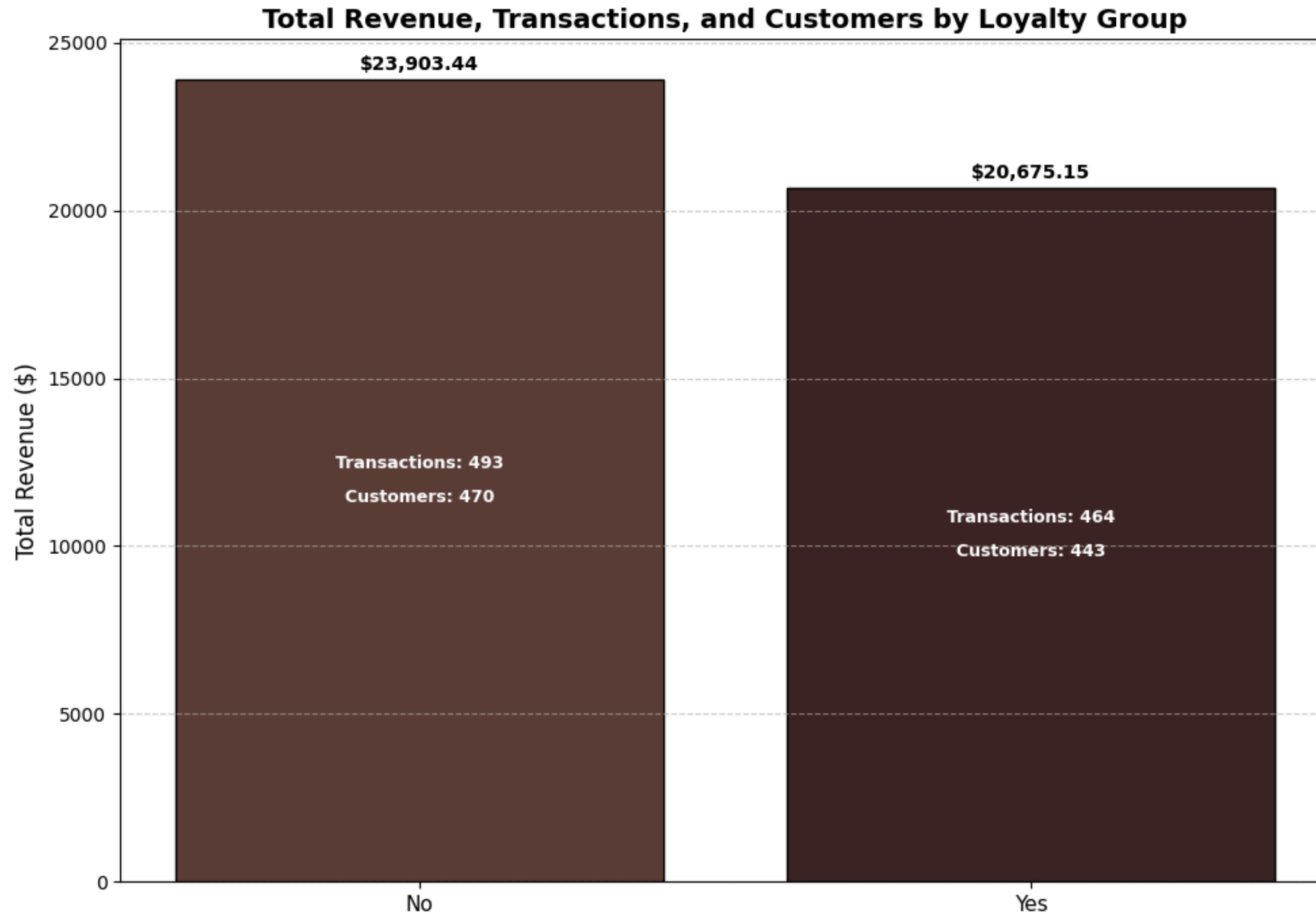
- **Margin Neutrality:** Loyalty members are not necessarily purchasing higher-margin items more frequently than regular customers.
- **Discount Offsets:** It is possible that any increase in spending by loyalty members is being neutralized by the discounts or rewards redeemed through the program.
- **Performance Gap:** The fact that loyalty members are numerically (though not significantly) less profitable suggests the program is currently a cost center rather than a direct profit driver per visit.

4. Analytical Recommendations & Program Optimization

To improve the effectiveness of the loyalty initiative, the following data-driven steps are recommended:

- **Shift to Lifetime Value (CLV) Analysis:** Evaluate whether loyalty members visit more frequently or remain customers longer. A program can be successful if it increases "Total Annual Profit" per customer, even if "Per-Transaction Profit" remains flat.
- **Granular Segmentation:** Audit loyalty tiers. Determine if "Gold" or high-tier members exhibit higher margins than "Entry-level" members, which may be masked in the aggregate data.
- **Cost-Benefit Audit:** Calculate the total administrative and discount costs of the program. If margins are neutral and visit frequency isn't significantly higher, the program's ROI should be formally questioned.
- **Targeted A/B Testing:** Experiment with "Margin-Positive" rewards. Instead of flat discounts, offer rewards for high-margin purchases (e.g., "Double points on Liberica roasts") to actively steer behavior toward more profitable products.
- **Qualitative Research:** Use customer surveys to determine if the program drives brand advocacy or referrals, which are valuable metrics not captured by transaction profit margins alone.

Total Revenue by Loyalty Group



Revenue and Engagement by Loyalty Group

1. Comparative Performance Metrics

The revenue audit reveals that the non-loyalty segment currently outpaces the loyalty segment in both total volume and individual transaction value, despite having a similar customer base size.

Metric	Non-Loyalty Customers	Loyalty Customers
Total Revenue	\$23,903.45	\$20,675.15
Unique Transactions	493	464
Unique Customers	470	443
Avg. Transaction Value	\$48.49	\$44.56

2. Analytical Drivers of the Revenue Gap

The \$3,228 revenue lead held by non-loyalty customers is driven primarily by Average Transaction Value (ATV) rather than foot traffic.

- **Transaction Quality vs. Quantity:** Non-loyalty customers spend approximately \$3.93 more per visit. This suggests they may be purchasing premium varieties (like Liberica) or larger bag sizes without the "safety net" of loyalty discounts.
- **Incentive Misalignment:** The loyalty program may currently reward frequency (number of visits) over depth (basket size). If rewards are triggered by the number of transactions, members may be splitting large orders into smaller, separate purchases to reach rewards faster, artificially lowering the ATV.
- **Acquisition Bias:** High initial spend by new, non-loyal customers may be inflating the non-loyalty totals. If these "big spenders" do not see the value in joining the loyalty program, the shop misses the opportunity to secure their long-term lifetime value.

3. Strategic Implications for Revenue Growth

This distribution indicates that the loyalty program is currently a retention tool rather than a revenue accelerator.

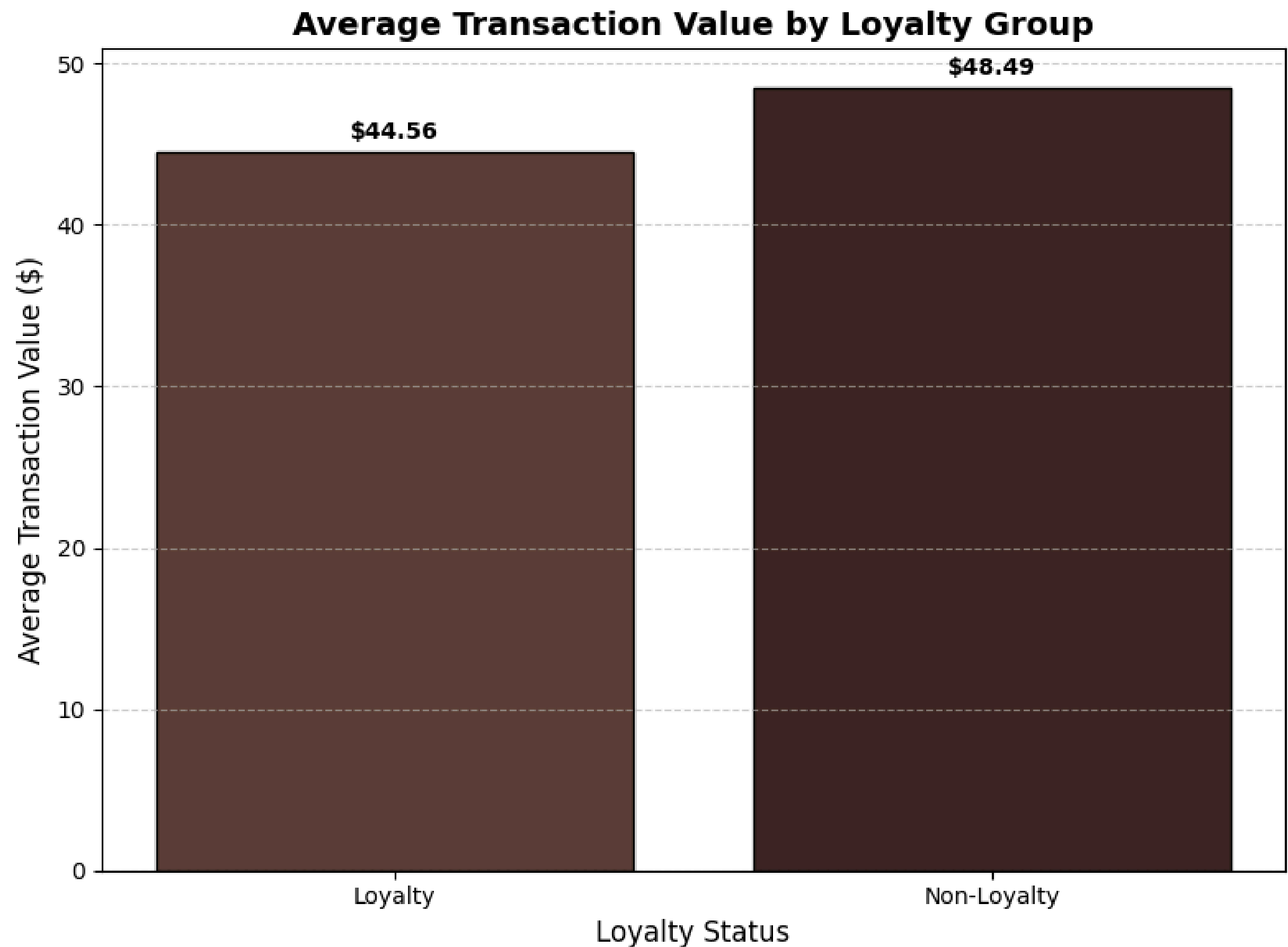
- **ATV Optimization:** The business must bridge the \$4.00 gap between loyalty and non-loyalty spending. Replicating the "high-spend" behavior of non-members within the loyalty group is the most immediate path to growth.
- **Segment Transition:** There is a high-value pocket of 470 non-loyalty customers. Converting the top 10% of these high-spenders into loyalty members—without diluting their spending through flat discounts—is a priority.

4. Targeted Action Plan

To rebalance the revenue distribution and optimize the loyalty program, the following steps are recommended:

- **Implement Spending Tiers:** Move away from a flat "points per visit" model. Introduce Spend-Based Tiers (e.g., Bronze, Gold, Platinum) where the rewards increase only as the transaction value grows.
- **High-Value "Upsell" Incentives:** Offer loyalty members exclusive access to limited-edition "Micro-lot" roasts or 2.5kg bulk options at a member-only rate that still maintains a high ATV.
- **A/B Testing Rewards:** Run a controlled test where Group A receives a discount on their next visit, while Group B receives a bonus for hitting a spending threshold in their current visit. Measure which drives higher immediate revenue.
- **Non-Loyalty "Big Spender" Conversion:** Identify non-members who spend over \$60 per transaction and offer a "Premium Membership" invitation. This ensures that the shop's most profitable customers are the ones being rewarded, rather than just the most frequent.

Average Transaction Value by Loyalty Group



Average Transaction Value (ATV) by Loyalty Group

1. Comparative Performance Overview

The analysis of transaction data reveals a counter intuitive spending pattern: customers who are not enrolled in the loyalty program are outspending enrolled members on a per-visit basis.

- **Non-Loyalty Customers:** Average Transaction Value of \$48.49.
- **Loyalty Customers:** Average Transaction Value of \$44.56.
- **Variance:** Non-loyalty customers spend \$3.93 more per transaction (approx. 8.8% higher).

2. Diagnostic Analysis of Influencing Factors

The \$3.93 gap suggests that while the loyalty program successfully drives retention, it is failing to drive basket depth. Key drivers include:

- **Frequency vs. Volume Optimization:** Loyalty members may be "gaming" the system by splitting large orders into smaller, more frequent transactions to accumulate visit-based points faster.
- **Specialty Bias in Non-Members:** Occasional visitors (non-members) often treat the shop as a destination for high-ticket "stock-up" items or premium gifts, whereas loyalty members treat it as a daily utility, purchasing smaller, routine items.
- **Discount Dilution:** Loyalty members are more likely to redeem points or coupons, which mathematically reduces the final transaction value recorded at the point of sale.

3. Strategic Implications for Growth

The current data indicates that the loyalty program is a frequency tool rather than a revenue accelerator. This presents a "Lost Revenue Opportunity":

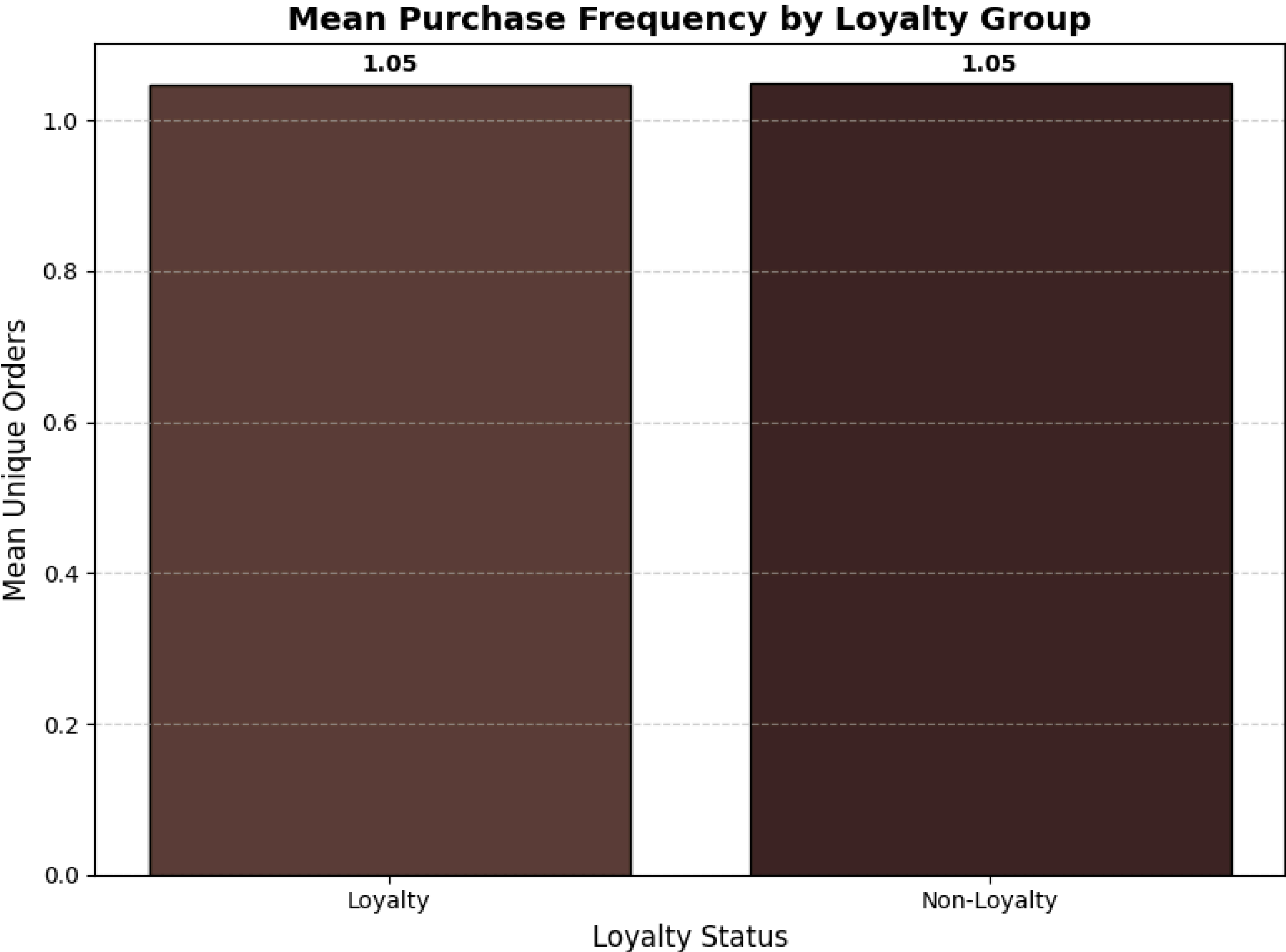
- **Basket Undersizing:** The business is failing to leverage its most "engaged" customers to drive higher-margin, high-volume sales.
- **Value Proposition Misalignment:** The program currently rewards "showing up" rather than "buying more."

4. Targeted Recommendations to Increase ATV

To bridge the gap and incentivize higher spending across both segments, the shop should pivot toward Spend-Centric Incentivization:

- **Tiered "Spend-to-Earn" Rewards:** Revamp the points engine. Instead of "1 point per visit," implement a tiered system where a \$60 transaction earns 3x bonus points, effectively encouraging loyalty members to consolidate their smaller purchases.
- **Smart Bundling and Cross-Selling:** Implement "Point-of-Sale" (POS) triggers. If a loyalty member is purchasing a standard roast, the POS should prompt staff to offer a high-margin companion product (e.g., premium filters or a specific pastry) with a "member-exclusive" bundle price.
- **"Threshold" Rewards:** Introduce a "Spend \$50, Get \$5" mechanic. This directly targets the current loyalty average of \$44.56, creating a clear psychological incentive for the member to add one more item to their basket to reach the \$50 threshold.
- **Premium Variety Promotion:** Actively market the higher-margin Liberica or Excelsa varieties to loyalty members. Since they already trust the brand, they are the ideal segment to "up-sell" into premium specialty roasts that naturally carry a higher unit price.
- **Data-Driven Personalization:** Use purchase history to identify "regular" items and offer a discount on a complementary premium item (e.g., "Since you love Arabica, take 20% off our limited-edition 1kg Liberica bag this week").

Mean Purchase Frequency by Loyalty Group



Purchase Frequency and Customer Retention

1. Comparative Frequency Metrics

The analysis reveals a surprising lack of differentiation in purchasing behavior between loyalty and non-loyalty segments. Despite the administrative overhead of maintaining a loyalty database, member behavior currently mirrors that of transient customers.

Metric	Loyalty Customers (\$n=443\$)	Non-Loyalty Customers (\$n=470\$)
Mean Orders	\$1.05	\$1.05
Median Orders	1	1
Std. Deviation	0.32	0.33
Max Orders	\$5.00	\$5.00

2. Diagnostic Findings: The "One-Time Buyer" Trap

The statistical data points to a high-volume, low-retention business model. With a median of 1.00 across both groups, the vast majority of the customer base—regardless of loyalty status—fails to return for a second transaction within the monitored timeframe.

- **Inert Loyalty Program:** The current program serves as a "passive database" rather than an "active motivator." If enrollment does not correlate with increased frequency, the rewards are either not visible enough or not valuable enough to influence habit formation.
- **Transient Foot Traffic:** The identical mean of 1.05 suggests the shop may rely heavily on its physical location and "walk-in" convenience rather than brand preference.
- **The "Exception" Customers:** While the range extends to 5 orders, these represent statistical outliers. The business is currently powered by a constant influx of new customers (acquisition) rather than the steady spend of a "regular" base (retention).

3. Strategic Implications for Business Health

- **High Acquisition Costs:** Relying on single-order customers is expensive. Every dollar spent on marketing or location is currently being "burnt" on a single transaction rather than being amortized over a long-term Customer Lifetime Value (CLV).
- **Loyalty Underperformance:** From a data science perspective, the loyalty program is currently failing its primary KPI: Inter-purchase Interval (IPI) Reduction. It is not shortening the time between visits or increasing the total visit count.

4. Targeted Retention Strategies

To convert one-time visitors into high-frequency "regulars," the shop should transition to an active-engagement model:

- **"Second-Visit" Automation:** Implement a "Next-Visit" trigger. If a new loyalty member doesn't return within 14 days, automate a personalized "We Miss You" offer (e.g., a 20% discount on their favorite variety) to break the single-purchase cycle.
- **Gamified Tiered Rewards:** Replace flat points with a Streak-Based System. For example: "Visit 3 times this month to unlock Gold Status." This creates a psychological "sunk cost" that encourages the customer to return to maintain their standing.
- **Member-Only "Early Access":** Distinguish the experience. Offer loyalty members a "First Taste" of seasonal roasts (like limited-run Liberica) 48 hours before the general public.
- **The "Third Purchase" Benchmark:** Statistical models often show that if a customer reaches a 3rd purchase, their churn probability drops significantly. Neo Coffee Shop should focus all loyalty incentives on reaching that 3rd transaction milestone.

Strategic Business Review

1. Product & Margin Optimization

The data confirms that profitability is highly segmented. While the business is growing, the "unit economics" vary significantly across the catalog.

- ***The Profit Leaders:*** Liberica is the primary margin driver. Future inventory should prioritize this variety to maximize net returns.
- ***Packaging Dynamics:*** The slightly higher margins on 0.2kg and 0.5kg bags suggest a "premium on freshness." Large-format (2.5kg) bags are likely over-discounted, as the volume gain does not currently offset the margin compression.

2. Financial Trajectory & Growth Stability

The time-series analysis provides a high-confidence outlook for stakeholders.

- ***Consistent Expansion:*** Cumulative growth is near-linear, indicating a stable market fit and reliable customer acquisition.
- ***Volatility Management:*** Monthly peaks and troughs suggest the business is susceptible to external cycles (seasonality). Strategic intervention is required to "fill the valleys" during traditionally slower months like May and August.

3. Loyalty Program Critical Audit

The most significant finding is the Neutral Impact of the current loyalty program.

- ***Statistical Parity:*** Inferential testing (Welch's t-test) proves there is no significant difference in profit margins between members and non-members (\$p=0.137\$).
- ***The Spend Paradox:*** Non-loyalty customers currently hold a higher Average Transaction Value (\$48.49 vs. \$44.56).
- ***The Retention Gap:*** Both groups share a mean purchase frequency of 1.05, indicating that the "Loyalty" label does not currently translate into "Repeat" behavior.

Strategic Pillar	Immediate Action Item	Expected Outcome
Pricing	Reduce bulk discounts on 2.5kg bags or introduce "Member-Only" value-adds for bulk.	Stabilize profit margins across all bag sizes.
Loyalty Revamp	Transition from visit-based rewards to Spend-Based Tiers (e.g., Bronze to Platinum).	Increase Average Transaction Value (ATV) for members.
Retention	Automate a "Second-Visit" discount via CRM for all first-time buyers within 14 days.	Break the "Single-Order" pattern (1.05 frequency).
Seasonality	Launch "Cold-Brew/Iced" marketing campaigns in May and August.	Smooth out intra-year revenue fluctuations.

Analytical Maturity Conclusion

Neo Coffee Shop is a financially healthy business with a strong growth foundation but an under-optimized customer engagement strategy. The transition from a passive loyalty model (tracking transactions) to an active behavior-steering model (incentivizing specific spend thresholds) will be the primary lever for accelerating 2026 profitability.